Routes That Fit You

A Personalised Multi-Objective Walking Route Planner with Learned Search Heuristics

CS F407 Artificial Intelligence - Project Report

Team Wayfinder · Track: Application (Search & Problem Solving + Applications of Deep Learning)

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Repository: https://github.com/ChaitanyaK07/WayFinder (https://github.com/ChaitanyaK07/WayFinder)

Abstract

We built a walking route planner for the Jubilee Hills / Banjara Hills area of Hyderabad in which the user, not the system, decides what matters. Five sliders weight walking time, junction load, traffic exposure, canopy (tree-cover proximity) and climb. The planner returns the best route under those weights on a turn-aware graph of 13,246 nodes, 34,708 directed edges and 103,384 turn transitions built from OpenStreetMap, ESA WorldCover and the Copernicus GLO-30 DEM.

We first measured the problem the proposal set out to study. As weight moves away from walking time, the admissible distance heuristic weakens: A* expansions rise from 35 % to 81 % of Dijkstra's. We then trained a preference-conditioned neural heuristic on 8,000,000 exact labels from 8,000 backward Dijkstra runs. It is very accurate: the mean ratio of predicted to true cost-to-go is 1.02, against 0.17 for the admissible heuristic, and it is equally accurate on weight vectors never seen in training. Used directly in A*, it expands 11× fewer states than A* with the admissible heuristic, but without any guarantee: routes are 4.4 % worse on average and up to 51 % worse.

Inside Focal Search, which guarantees a route within W × the optimum, the network does not pay off. All 7,000 bounded runs respected their guarantee. However, the learned guide saves only 15 % of expansions against A* at W = 1.5, about the same as weighted A* and a tuned classical guide. Even a perfect guide saves only 18 %. We show why: FOCAL admits a state only within W × f_min, and f_min is driven by the weak admissible heuristic. We measure this with a per-query "focal floor". Following the proposal's Week-4 rule, the headline therefore falls back to Focal Search with a tuned classical guide.

The delivered application returns the exact best route by default. This was verified on 147 random trips, with 0 violations of slider monotonicity in 735 checks. The human evaluation is fully prepared but was not run; Section 10 explains why and what can be concluded without it.

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1. Introduction

1.1 Motivation

Mainstream route planners answer one question: what is the fastest way there? Many walking trips are not about that. A student with a heavy bag will accept a few extra minutes to avoid a staircase. Someone recovering from a knee injury routes around gradients. In a Hyderabad summer, walking under tree cover is worth a detour. These preferences differ from person to person and from trip to trip, and no widely available tool lets a person state them.

1.2 Why this is technically hard

A* beats Dijkstra by using a heuristic: an estimate of the remaining cost that pushes the search toward the goal. For plain travel time, straight-line distance is a sound estimate. It never overestimates (admissibility), which is what makes A* return optimal routes. Once the cost also includes junction penalties, traffic exposure, canopy and climb, a distance-based estimate stays admissible but becomes a weak guide, and A* degenerates toward Dijkstra. The stronger the user's non-distance preferences, the worse this gets — exactly when personalisation matters most.

1.3 What we did

1. A **turn-aware, five-objective cost model** on a real street graph, with every feature labelled *measured*, *derived* or *proxy* (Section 3–4).
2. An **admissible and consistent heuristic** whose consistency is guaranteed by construction, and checked on every edge (Section 4.3).
3. The **problem stated as a measurement**: expansions against non-distance weight (Section 9.1).
4. A **preference-conditioned learned heuristic**: one network for every slider setting, trained on exact labels from backward Dijkstra (Section 5).
5. **Focal Search** that uses the learned heuristic for guidance and the admissible heuristic for a provable `W ×` bound, with the bound verified for W = 1.0 … 2.0 against exact optima (Sections 6, 9.3).
6. An **evaluation** on 200 frozen queries with three training seeds, heuristic-quality analysis and generalisation to held-out preferences (Section 9).
7. A **routing service and web interface** with five sliders, verified to return the exact best route (Sections 7, 9.9).

The project makes no novelty claim. Its contribution is a ground-truth-verified evaluation of preference-conditioned learned heuristics and Focal Search, including a negative result and the measurement that explains it.

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2. Background

Best-first search. Dijkstra's algorithm expands states in order of path cost `g`. A* (Hart, Nilsson and Raphael, 1968) orders them by `f = g + h`, where `h` estimates the remaining cost. If `h` never overestimates (*admissible*), A* returns an optimal path. If `h(u) ≤ c(u, v) + h(v)` for every edge (*consistent*), A* also never needs to re-expand a state.

Bounded suboptimality. Weighted A* orders by `g + W·h`. With a consistent `h` it returns a path of cost at most `W ×` optimal. This guarantee depends on `h` being admissible; it does not hold for a learned, overestimating `h`.

Focal Search (Pearl and Kim, 1982) separates the two jobs of a heuristic. OPEN is ordered by a safe `f = g + h_adm`. FOCAL is the subset of OPEN whose `f` is within `W ×` the current minimum. A second heuristic — which may overestimate — decides which FOCAL state to expand next. The bound comes from the safe heuristic and the guidance from the second one.

Learned heuristics. A network can learn the cost-to-go from examples. In our setting the labels are exact: one backward Dijkstra run from a target gives the true optimal cost from every node. Conditioning the network on the user's weight vector lets one model serve every slider setting.

node2vec (Grover and Leskovec, 2016) learns node embeddings from biased second-order random walks with a skip-gram objective. Its return parameter `p` and in-out parameter `q` move the walks between breadth-first and depth-first exploration.

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3. Study area and data

3.1 Choosing the area: the coverage audit

Before writing the cost model we audited candidate areas (`results/audit/`, and the proposal's Section 3). Around the BITS campus, OSM mapping was too sparse: no street names or stairways. Miyapur passed most checks, but canopy preference barely changed routes: 92 % overlap with the fastest route at weight 0.8 (`results/audit/comparison.csv`).

Jubilee Hills gave a working signal on every feature. On the final 7 × 7 km graph, a preference weight of 0.8:

- cut traffic exposure by 29 %, climb by 26 % and turns by 57 %;
- more than doubled canopy along the route;
- in each case at a mean detour of 12 % or less (`results/audit/jubilee_3500_verdict.csv`).

3.2 The graph

Property	Value
Centre, extent	17.4220 N, 78.4200 E; 7 × 7 km; UTM zone 44N (EPSG:32644)
Nodes / directed edges	13,246 / 34,708 (largest strongly connected component = 100 % of nodes)
Turn-aware states / transitions	34,708 / 103,384
Network length (both directions)	2,075 km
Elevation relief	121 m
Stairway edges	60

Source: `data/processed/graph_summary.json`. All data is open: OpenStreetMap (ODbL), ESA WorldCover 2021 (CC BY 4.0) and Copernicus GLO-30 DEM. We used no purchased data, API keys or scraping.

3.3 Features

Every feature is a per-metre score in [0, 1], except junction load, which is priced per turn. We label each one honestly as measured, derived or proxy.

Feature	Source	Status	Definition	Mean (length-weighted; junction: per transition)
Walking time	OSM highway class, length	measured	`0.5 / speed`; 1.3 m/s on footways … 0.5 m/s on steps	0.42
Junction load	node degree, way geometry	derived, exact	turn angle from bearings (<30° → 0, <75° → 0.3, <150° → 1, U-turn → 3), +0.2 at nodes with ≥ 3 neighbours	1.61
Traffic exposure	OSM highway class	proxy	ordinal rank: trunk/primary 1.0 … footway 0.0	0.39
Canopy proximity	ESA WorldCover 2021, 10 m	proxy	`1 − tree-cover share` in a 15 m buffer	0.91
Climb	Copernicus GLO-30 DEM; `highway=steps`	derived, reliable	uphill grade clipped at 15 %, scaled to [0, 1]; stairs = 1	0.12

Source: `results/feature_summary.csv`.

Why "canopy", not "shade". Real shade modelling needs building heights and the sun's position. Building-height tags in Hyderabad OSM are too sparse for that. What we can defensibly compute is proximity to tree cover. We call it canopy proximity and label it a proxy: it correlates with shade but does not measure it. Likewise, OSM carries no traffic counts, so traffic exposure is an ordinal ranking of road

classes.

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4. Problem formulation

4.1 Cost model

For a weight vector `w = (w_time, w_junction, w_traffic, w_canopy, w_climb)` with every slider in [0, 1]:

```
rho(e, w)          = 0.1 + Σ_k w_k · f_k(e)            k ∈ {time, traffic, canopy, climb}
c(e_in, e_out, w)  = length(e_out) · rho(e_out, w)  +  w_junction · 30 m · turn_penalty(e_in, e_out)
```

The base term 0.1 keeps every metre's cost above zero for every weight vector. Without it, the admissible heuristic below would collapse to zero whenever all non-distance weights are zero on some edge. A penalty-1 turn costs as much as 30 m of walking at full junction weight.

4.2 Turn-aware state space

A search state is the pair *(node, incoming edge)* — equivalently, the edge just walked. Transition `e → e'` exists when `e'` leaves the node `e` enters, and it costs `c(e, e', w)`. The origin is a virtual state with no incoming edge, so the first edge carries no turn cost. A query is solved when a state whose head is the destination is expanded. U-turns are allowed (penalty 3) and self-loops (cul-de-sacs) are kept, so turning around is priced, not forbidden.

4.3 The admissible heuristic, and why it is consistent

```
h_adm(v, t, w) = rho_min(w) · s · haversine(v, t),      rho_min(w) = min_e rho(e, w)
```

OSMnx measured edge lengths as sums of great-circle segments on a sphere of radius 6,371,009 m. We use haversine on the same sphere. `s` is the smallest `length(e) / haversine(u, v)` over all edges, times $(1 - 10^{-9})$ for rounding; on our graph `s = 0.999999997`. Then for every edge `e = (u, v)` and any incoming state:

```
h_adm(u) ≤ rho_min · s · (haversine(u, v) + haversine(v, t))    triangle inequality
         ≤ length(e) · rho(e, w) + h_adm(v)                        definition of s, rho_min
         ≤ c(e_in, e, w) + h_adm(v)                                turn penalties ≥ 0
```

Since `h_adm(t) = 0`, `h_adm` is consistent and therefore admissible. The test suite checks this inequality on every edge of the real graph, and the label pipeline checked `h_adm ≤ h*` on all 8,500,000 labelled samples: 0 violations (`data/labels/labels_summary.json`).

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5. The learned heuristic

5.1 Exact labels from backward Dijkstra

For a target `t` and weight vector `w`, one Dijkstra run on the *reversed* state graph gives `D(e)`: the optimal cost from every state to `t`. The node-level label is the optimal cost of a route that *starts* at `v`:

```
H(v) = min over edges e' leaving v of [ cost(e') + D(e') ],      H(t) = 0
```

Turn penalties are non-negative, so `H(head(e)) ≤ D(e)`. A node-level heuristic trained on `H` therefore targets a lower bound of the true state cost-to-go.

The 8,000 backward runs (40 weight vectors × 200 targets) use SciPy's compiled Dijkstra and take about one to two minutes on one core. The test suite cross-checks them against a pure-Python reference implementation. From each run we keep 1,000 uniformly sampled nodes; every label is exact, and sampling only bounds the dataset size.

The published dataset (`data/labels/`, described in `DATASHEET.md`):

Folder	Weight vectors	Targets	Runs	Samples
`train_weights/`	40 training	all 200	8,000	8,000,000
`heldout_weights/`	10 held-out	50 (val + test)	500	500,000

Each folder holds one compressed shard per weight vector (`w00.npz` …, about 3.4 MB each) and an `index.json` with the weight vectors and a SHA-256 per shard, which is checked whenever the labels are loaded.

5.2 Weight vectors and splits

The weight vectors are frozen once in `data/splits/weights.json`: 40 for training and 10 held out. Each slider is drawn uniformly from [0, 1] and switched off with probability 0.3, to cover sparse preferences.

The 200 label targets are frozen in `data/splits/targets.json` and split **by target node**: 150 train, 25 validation, 25 test. The model therefore never sees a test destination, and never sees a held-out weight vector while training or choosing hyperparameters.

5.3 Inputs: node2vec embeddings

We implemented node2vec from scratch. It uses second-order biased walks (`p = 1`, `q = 0.5`; 10 walks of length 40 per node, 132,460 walks in total), vectorised over all walks at once. Training is skip-gram with 5 negative samples (window 5) for 2 epochs, with the loss falling from 0.220 to 0.095, giving 32-dimensional embeddings (`data/processed/node2vec.json`).

5.4 Network and ratio parameterisation

```
x       = [ node2vec(v), node2vec(t), w, (Δx, Δy, ‖Δ‖) / 1 km ]   32 + 32 + 5 + 3 = 72 inputs
r_hat   = 1 + softplus( MLP(x) )                                  MLP: 72 → 256 → 256 → 1, ReLU
h_learn = r_hat · h_adm(v, t, w)
```

The network predicts the *ratio* of the true cost to `h_adm`, not the cost itself. Since `r_hat ≥ 1`, `h_learn` is never less informed than the classical heuristic, and the target ratio `h*/h_adm` is scale-free across weight vectors. The loss is the mean squared error of `log r_hat` against `log(h*/h_adm)`.

5.5 Training

We used Adam with batches of 4,096, halved the learning rate when validation stalled, and stopped early with patience 3, checkpointing the best epoch. A hyperparameter pass over hidden size × learning rate, on 1.5 M samples for 3 epochs, chose 256 × 256 at lr = 0.003:

Hidden	Learning rate	Validation MSE (log ratio)
256, 256	0.003	**0.0155**
128, 128	0.003	0.0165

Hidden	Learning rate	Validation MSE (log ratio)
128, 128	0.001	0.0181
256, 256	0.001	0.0182

Three seeds were then trained on all 6,000,000 training-target samples. Their best validation MSEs were 0.0169, 0.0138 and 0.0161, at epochs 1, 10 and 2 (`results/training/`). The training loss keeps falling while the validation loss stalls early, which shows that generalising to *unseen destinations* is the hard part with only 150 training targets.



5.6 Batched inference with cached embeddings

The first layer is linear, so W_{1x} splits into a node term, a target term, a weight term and an offset term. The node and target terms are precomputed once for every node (the cached embeddings), and the weight and target terms once per query. Only the 3-number offset and the remaining layers run per node, in numpy.

When the search first meets a node, the whole 250 m spatial tile containing it is inferred in one batch, so only the region the search actually touches is computed. A test checks that this numpy port agrees with PyTorch.

5.7 Tuned classical fallback

The proposal's Week-4 fallback needs a strong *classical* guide. We fitted $h_{tuned} = \kappa(w) \cdot h_{adm}$, with $\log \kappa$ linear in w, by least squares on the training labels. It uses no node information, so it is a tuned version of the classical heuristic rather than a learned one (`models/tuned_classical.json`).

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6. Search algorithms

All searches run on the turn-aware state graph and count states **expanded** (taken off OPEN; the origin counts as one) and **generated** (inserted into OPEN, new or with a better g).

Dijkstra, A*, weighted A*. One best-first search ordered by $g + W \cdot h$, breaking ties toward larger g. With the consistent h_{adm}, A* never re-expands and weighted A* keeps its $W\times$ bound without re-expansions.

Focal Search. OPEN is ordered by $f = g + h_{adm}$. FOCAL holds OPEN states with $f \le W \cdot f_{min}$. The next state is the FOCAL state with the smallest $g + h_{learn}$. The implementation keeps three heaps with lazy deletion and per-state version stamps. When f_{min} rises, states move from a waiting heap into FOCAL, and a FOCAL entry found above the current bound is moved back. Re-opening is on.

Why the bound holds. With re-opening, some state on an optimal path is always on OPEN with its optimal g, so $f_{min} \le C^*$. A goal is only ever taken from FOCAL, where $g = f \le W \cdot f_{min} \le W \cdot C^*$.

Focal floor. For every query and W we also count the states with `g* + h_adm < C*/W`, using a forward Dijkstra. Focal Search stops only once `f_min ≥ C*/W`, and with re-opening each such state keeps a lower-`f` state on OPEN until it is expanded. So **every** Focal Search must expand all of them, whatever its guide. The test suite checks this lower bound for perfect and adversarial guides.

The tests also run Focal Search with a random, wildly inadmissible guide at W ∈ {1.0, 1.1, 1.5, 2.0}: the bound always holds, and W = 1.0 is always exactly optimal.

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7. System

- **Pipeline** (`python -m wayfinder <step>`): preprocess → splits → benchmark → labels → node2vec → train → evaluate → report → study-prepare / study-analyze. Each step only reads and writes files. Splits and benchmark are frozen and refuse to be overwritten. All settings live in `configs/default.yaml`.
- **Routing service** (FastAPI): `POST /api/route` takes an origin, a destination and the five weights, snaps the points to the network, searches, and returns the route geometry, its metrics, search statistics and the fastest route for comparison. Route pricing is cached per weight vector, behind a lock, because requests arrive on a thread pool.
- **Web interface** (Leaflet): click a start and a destination, then drag five horizontal sliders or pick a preset. The route updates as you drag, and only the newest request may draw, so a slow earlier reply cannot overwrite it. **By default the search is exact (A*)**: the displayed route is the best possible route for the chosen sliders. Focal Search and weighted A* are opt-in modes labelled "within W × best".
- **Tests:** 87 tests, run on a synthetic fixture graph and on the real graph (Section 12).

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8. Experimental setup

- **Benchmark:** 200 queries frozen with a fixed seed (`data/benchmark/queries.csv`, with a SHA-256 checked on every load). Origin and destination are at least 700 m apart (median 3.4 km, max 7.7 km), and no destination is a label target. 100 queries use a training weight vector and 100 a held-out one.
- **Ground truth:** the exact optimum of every query, from backward Dijkstra.
- **Conditions:**
  - Dijkstra; A*(h_adm); weighted A* at W = 1.25, 1.5, 2;
  - A*(h*) (oracle);
  - A*(h_learn) (no bound);
  - Focal Search guided by h_learn, by h_tuned and by h* (oracle), each at W = 1.0, 1.1, 1.25, 1.5, 1.75, 2.0.
  - Learned conditions are run with all three seeds (7,800 runs in total).
- **Metrics:**
  - states expanded and generated;
  - latency (graph pricing + heuristic preparation + search, one process);
  - suboptimality (cost / exact optimum) and bound violations;
  - heuristic admissibility violations, consistency violations and informedness `h/h*`.
- **Hardware:** a laptop CPU, single-threaded Python, no GPU.

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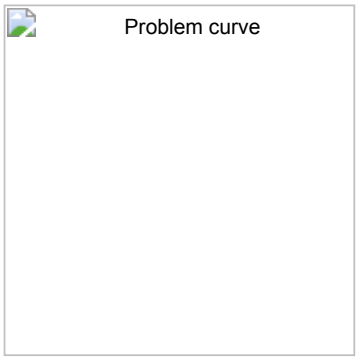
9. Results

9.1 The problem, stated as a measurement

The λ family `w = (1 − λ, λ/4, λ/4, λ/4, λ/4)` moves weight from walking time to the four other sliders. This is on the 200 frozen origin–destination pairs (`results/problem_curve.csv`):

λ	rho_min	Dijkstra	A*(h_adm)	A*(h_adm) / Dijkstra	h_adm / h* at origin	A*(h*)
0.0	0.485	17,316	6,026	**0.35**	0.67	87
0.2	0.408	17,321	7,787	0.45	0.56	85
0.4	0.331	17,324	9,617	0.56	0.46	84
0.6	0.254	17,311	11,404	0.66	0.35	84
0.8	0.177	17,286	13,105	0.76	0.25	85
0.9	0.138	17,272	13,974	**0.81**	0.19	85

(mean states expanded per query)



As preferences move away from distance, `rho_min` falls, `h_adm` recovers less and less of the true cost (67 % → 19 %), and A* drifts toward Dijkstra. The perfect heuristic would need about 85 expansions throughout, so the headroom a better heuristic could in principle exploit grows from about 70× at λ = 0 to about 160× at λ = 0.9. The figure also shows learned Focal Search at W = 1.5: at low λ it expands *more* than A*, and it never comes close to the headroom. Section 9.8 explains why.

9.2 Search efficiency on the frozen benchmark

Medians over the 200 queries; learned conditions pool three seeds (`results/benchmark_summary.csv`):

Condition	W	Expanded (median)	IQR	Focal floor	Latency median / p95 (ms)	Mean / max suboptimality
Dijkstra	–	17,574	6,975 – 26,970	–	72 / 178	1 / 1
A*(h_adm)	1	13,168	5,398 – 22,265	–	63 / 183	1 / 1
Weighted A*	1.5	11,054	4,847 – 20,154	–	56 / 184	1.000 / 1.060
Weighted A*	2	9,586	3,701 – 17,817	–	51 / 176	1.002 / 1.062
Focal, tuned classical	1.5	11,029	4,616 – 19,134	4,877	109 / 379	1.000 / 1.030
Focal, tuned classical	2	8,812	3,749 – 16,921	2,110	90 / 395	1.001 / 1.053
Focal, learned	1.5	11,161	4,619 – 19,252	4,877	261 / 783	1.001 / 1.056
Focal, learned	2	8,552	3,555 – 17,619	2,110	204 / 726	1.002 / 1.108
Focal, perfect guide h* (oracle)	1.5	10,847	4,270 – 18,260	4,877	110 / 340	1 / 1
Focal, perfect guide h* (oracle)	2	8,049	2,820 – 15,530	2,110	76 / 304	1 / 1
A*(h_learn), no bound	–	**1,205**	402 – 2,351	–	**40** / 144	1.044 / 1.506

Condition	W	Expanded (median)	IQR	Focal floor	Latency median / p95 (ms)	Mean / max suboptimality
A*(h*) (oracle)	1	84	54 – 112	–	6 / 11	1 / 1

Per-seed medians for learned Focal at W = 1.5 are 11,285.5, 11,060 and 11,269, and for A*(h_learn) 1,340, 1,031.5 and 1,177.5. The seeds agree closely.



9.3 The suboptimality bound holds

Across all 7,000 bounded runs (every Focal condition, every W, three seeds, plus weighted A* and A*), **no route exceeded its guaranteed bound** (`results/bound_sweep.csv`). In practice the routes are far better than guaranteed. For learned Focal, the worst query was 1.006×, 1.033×, 1.056×, 1.056× and 1.108× the optimum at W = 1.1, 1.25, 1.5, 1.75 and 2.0, and the mean stayed within 0.2 % of optimal.



9.4 Heuristic quality

Averaged over the 200 benchmark queries, at 2,000 sampled nodes each (`results/heuristic_quality.csv`):

Heuristic	h / h* mean (median)	Overestimates h*	Mean overestimate	Consistency violations
h_adm	0.167 (0.168)	0 %	0 %	0 %
h_tuned	1.029 (1.038)	50.2 %	13.3 %	24.3 %
h_learn, seed 0	1.024 (1.014)	53.5 %	10.7 %	32.7 %
h_learn, seed 1	1.031 (1.019)	55.5 %	10.4 %	31.8 %
h_learn, seed 2	1.022 (1.009)	51.2 %	10.6 %	33.2 %



The learned heuristic recovers essentially the whole true cost: its median `h/h*` is within 2 % of 1. It overestimates about half the time, by about 10 % on average, which is why it needs Focal Search's safety net. Its spread is tighter than the tuned classical heuristic's. The tuned heuristic is surprisingly good *on average*, because one scalar κ(w) per weight vector already captures the typical detour ratio, but it is wider: its histogram has a long right tail.

9.5 Generalisation to held-out preferences

On label level, for unseen test targets (median over seeds; `results/training/label_metrics.csv`):

	MAE of log(h*/h_adm)	**h_learn / h***	**h_adm / h***
training weight vectors	0.100	1.041	0.211
held-out weight vectors	0.104	1.013	0.117

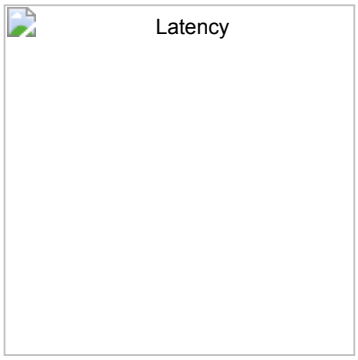
On search level, the median expanded states split by weight set (`results/generalisation.csv`): A*(h_learn) needs 1,068 on held-out weights against 1,340 on training weights. Learned Focal at W = 1.5 needs 11,726 against 11,065.

The network **generalises rather than memorises**: its accuracy on preferences it never saw matches the training preferences. The held-out queries are harder for the classical heuristic, whose informedness drops to 0.117, but not for the learned one.



9.6 Latency

All conditions run in single-threaded Python and are interactive (`results/plots/latency.png`). A*(h_adm) has a median of 63 ms. A*(h_learn) is the fastest practical option at 40 ms, because it expands 11× fewer states. Learned Focal is the slowest, at 261 ms median and 783 ms p95: it pays for network inference *and* expands as many states as A*.



9.7 Week-4 go / no-go

The proposal's gate asks whether the learned heuristic expands fewer nodes than the classical baseline. We applied it strictly: GO requires beating both A*(h_adm) and the tuned classical guide at the same W (`results/go_no_go.json`).

- Learned Focal against A*(h_adm) at W = 1.5: 11,161 vs 13,168, **15.2 % fewer** ✔
- Learned Focal against tuned Focal at W = 1.5: 11,161 vs 11,029 ✘
- Learned Focal against weighted A* at W = 1.5: 11,161 vs 11,054 ✘

Verdict: NO-GO. Following the proposal's fallback, the headline method is Focal Search guided by the tuned classical heuristic, and the rest of this section is the measurement of why the learned version did not pay off.

9.8 Why Focal Search cannot exploit a good heuristic

The learned heuristic is excellent (Section 9.4), and A* with it is 11× cheaper. Yet inside Focal Search it barely helps — and **neither does a perfect heuristic**. Focal Search guided by the true `h*` saves only 17.6 % at W = 1.5.

The reason is structural. A goal can only be taken from FOCAL once `f_min ≥ C*/W`, and `f_min` is the minimum of `g + h_adm` over OPEN. With `h_adm` recovering only 17 % of the true cost, the region of states with `g + h_adm < C*/W` is large, and every one of those states must be expanded before the search can stop.

At W = 1.5 this **focal floor** is 4,877 states (median). The observed expansions are above 10,800 for *every* guide, because the guide spends expansions on near-optimal states while `f_min` still has to be raised.

Two practical consequences follow:

- The guarantee of Focal Search is only as useful as its admissible heuristic. Better guidance cannot buy speed until `h_adm` itself is stronger, for example with ALT landmarks (Goldberg and Harrelson, 2005), which the proposal placed out of scope.
- Where a guarantee is not required, the learned heuristic in plain A* gives most of the possible gain. It cuts expansions 11×, keeps routes within 4.4 % of optimal on average, and reaches 51 % worse only in the worst query.

We also piloted a bound-preserving "incumbent" FOCAL policy, which switches to the `f_min` state once a goal has been generated. On eight benchmark queries it saved at most about 3 %, so we removed it: the extra expansions happen *before* the goal is ever generated.

9.9 The application returns the best route

`scripts/verify_routes.py` drew 147 random trips with random slider settings on the real graph. Every route the application returned by default equalled the exact Dijkstra optimum, both by its reported cost and when re-priced from its edges (**147/147**, worst ratio 1.000000000000).

For an exact optimiser, raising one slider cannot increase the route's total of what that slider penalises. We checked this for all five sliders on every trip: **0 violations in 735 checks**. Moving each slider from 0 to 1 changed the route in 106 (time), 141 (junction), 136 (traffic), 129 (canopy) and 115 (climb) of the 147 trips.

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10. Discussion and limitations

What worked.

- The cost model gives a real signal on every slider.
- The admissible heuristic is provably consistent.
- Exact labels make learning easy to evaluate.
- The preference-conditioned network is accurate and generalises to unseen preferences.
- Focal Search's bound held in every one of 7,000 runs.
- The application returns provably best routes interactively.

What did not. The proposal's headline — learned guidance making bounded-suboptimal search substantially faster — did not materialise. The cause is not the network but the search framework: Focal Search's effort is set by the admissible OPEN heuristic. We consider the focal-floor measurement, together with the perfect-guide control, the main technical finding of the project.

Threats to validity and limitations.

- *Proxies.* Traffic exposure is ranked from road class, not measured. Canopy is tree-cover proximity, not shade. The climb score inherits 30 m DEM noise; grades are clipped at 15 % for that reason.
- *One study area.* All results are on one 7 × 7 km graph. The coverage audit shows that sparse OSM areas such as the BITS campus would behave differently.
- *Generalisation to destinations.* With 150 training targets, validation loss stalls after a few epochs. More targets would likely tighten the learned heuristic further.
- *Latency* is measured in single-threaded Python. The relative order of conditions is what matters; absolute times would fall several-fold in a compiled implementation.
- *No human evaluation* yet (Section 10).

Future work.

- A stronger admissible bound for OPEN, such as ALT landmarks or the max of several admissible estimates, so that Focal Search can use its guide.
- Explicit Estimation Search, which uses an inadmissible estimate to decide *when* to raise `f_min`.
- Training against an admissible ceiling, to trade a little informedness for fewer overestimates.
- Running the user study.

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11. Reproducibility

Everything needed is in the repository. That includes the graph, frozen splits and benchmark, the label dataset, embeddings, trained models and every result file.

```
python -m pip install -r requirements-dev.txt && python -m pip install -e . --no-deps
python -m pytest                        # 87 tests: fixture graph + real graph
python scripts/verify_routes.py         # 147 random trips: exact routes, slider monotonicity
python -m wayfinder serve               # web app at http://127.0.0.1:8000
python -m wayfinder all                 # whole pipeline from the GraphML (~1.5 h on a laptop CPU)
```

Step	Time
Labels (8,000 runs)	1–2 min
node2vec	~4 min
Training (hyperparameter pass + 3 seeds)	25–40 min
Evaluation	40–50 min

Every number in Section 9 is copied from a file under `results/`. `python -m wayfinder report` regenerates `results/summary.md` and all figures from the raw result files.

Appendix A — Work division

Pair	Build phase (Weeks 1–4)	Second phase (Weeks 5–7)	Members
Graph	OSM extraction, coverage audit, UTM projection, turn-aware graph, five feature extractors, cost model	routing service (FastAPI), web interface with sliders	Anisha Potdar, Ananya Avunooru
Search	Dijkstra, A*, weighted A*, admissible heuristic, turn-aware expansion, node accounting, batched inference	Focal Search, W sweep, empirical bound verification	Srujana N S, Anushka Sahoo
Data	backward Dijkstra at scale, weight-vector sampling, label pipeline, dataset publication, node2vec	evaluation harness, heuristic-quality analysis, plots	Soujanya Lakshmi Tayyala, Chandana N
Model	network architecture, ratio parameterisation, training loop, hyperparameter pass, checkpointing	generalisation on held-out weights, user study	Vinny Polamarasetty, Chaitanya Srikar Kastury

Appendix B — AI-usage disclosure

A Part of implementation, tests, evaluation runs and doc writings were produced with the help of an AI assistant (Claude Code), as logged in `docs/ai_usage_log.md`. Every claim is backed by a result file and every algorithm by tests against exact optima. Each pair is responsible for being able to explain its components in the viva.

Appendix C — Main settings (`configs/default.yaml`)

Setting	Value
Weight vectors	40 training + 10 held-out; slider off with probability 0.3; seed 7
Label targets	200 (150 / 25 / 25 by target); 1,000 nodes kept per run
Benchmark	200 queries, ≥ 700 m apart, half on held-out weights; seed 2025
node2vec	32-d, p = 1, q = 0.5, 10 walks × 40 steps, window 5, 5 negatives, 2 epochs
Network	72 → 256 → 256 → 1, ReLU, `r_hat = 1 + softplus`
Training	Adam, lr 0.003, batch 4,096, ≤ 12 epochs, patience 3, seeds 0, 1, 2
Evaluation	Focal W ∈ {1.0, 1.1, 1.25, 1.5, 1.75, 2.0}; weighted A* W ∈ {1.25, 1.5, 2}; FOCAL key `g + h_learn`; headline W = 1.5

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